YOUR APP, THEIR DATA

Local-first with Evolu

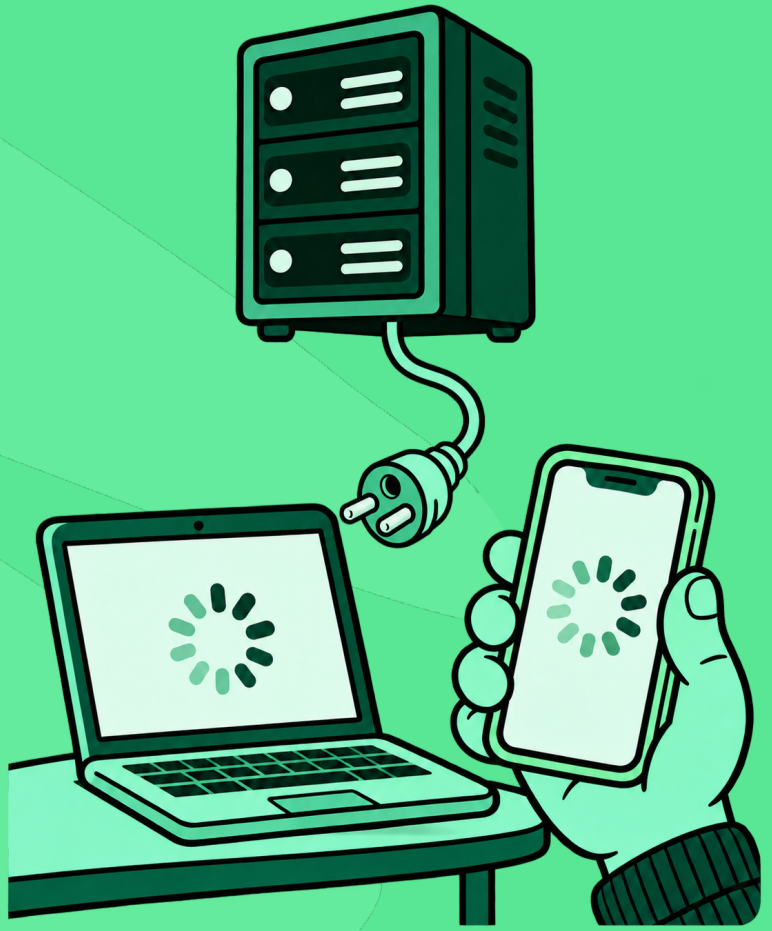
Matěj Kříž / Trezor



Your servers are gone

What can your users still do?

Read?
Modify?
Export?



TAKE CONTROL



Fee −2.2 BTC −$0.02

May 22, 2010 −10,000.99 BTC −$40.77

Sent BTC • 20:16

Papa John's — two pizzas 🍕

	−10,000 BTC	−$40.76
Fee	−0.99 BTC	−$0.01

April 16, 2010 +500 BTC +$2.30

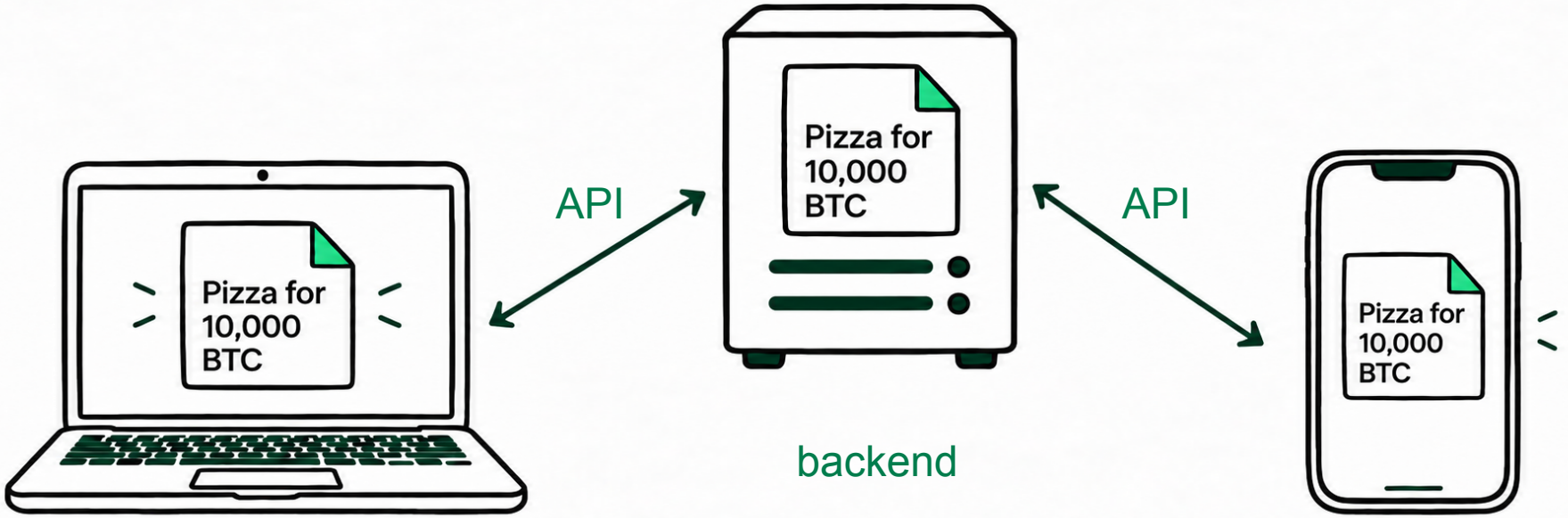
Received BTC • 23:11

vexl +500 BTC +$2.30





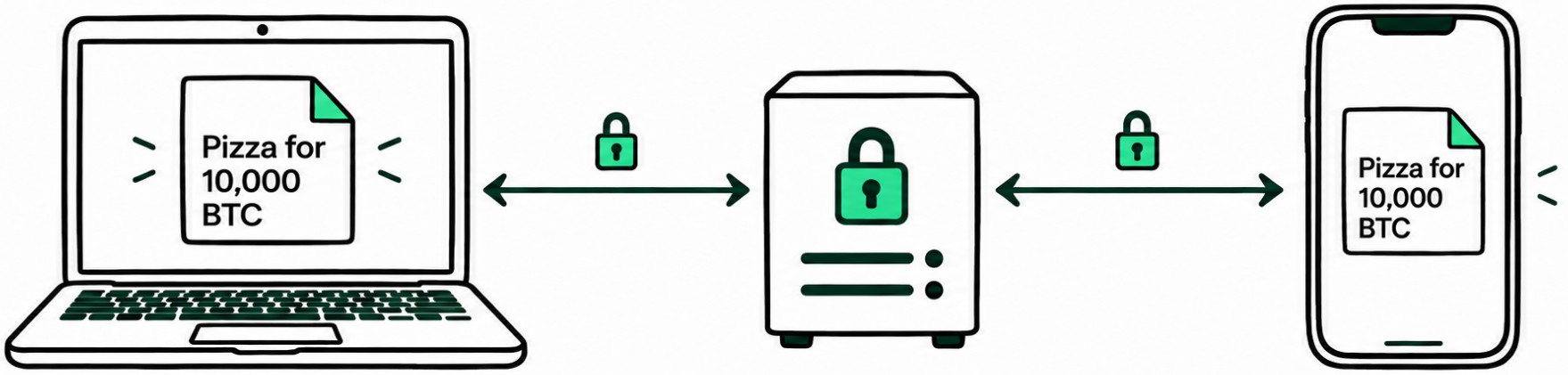
Cloud architecture



YOUR DATA IS NONE OF OUR BUSINESS



Local-first architecture

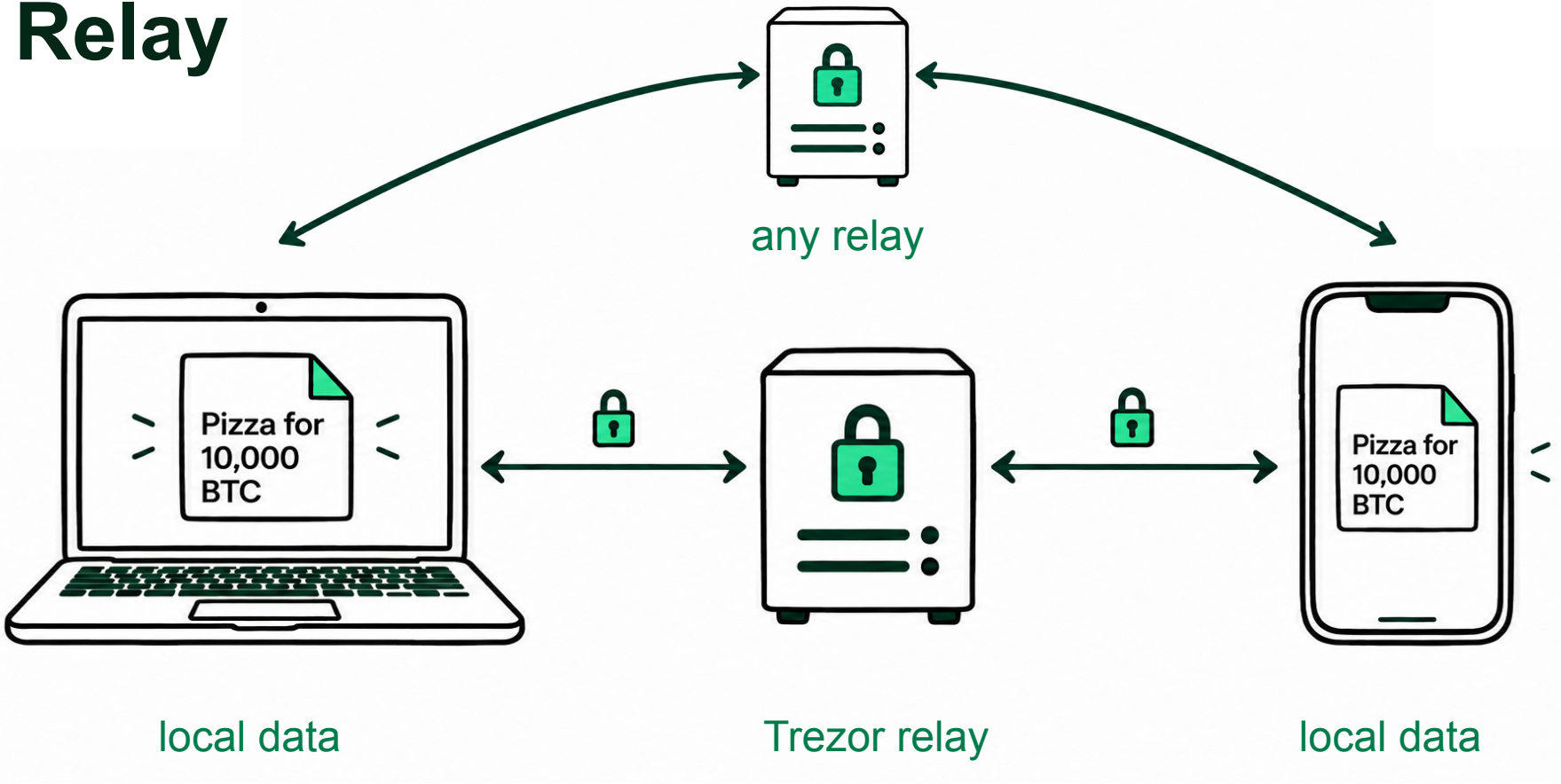


local data

Trezor relay
encrypted backup

local data

Relay


any relay
Pizza for 10,000 BTC
Pizza for 10,000 BTC
local data
Trezor relay
local data

SYNCING IS HARD



SYNCING ~~IS~~ WAS HARD





Local-first

You own your data, in spite of the cloud

Ink & Switch, 2019

7 IDEALS
FOR Local-First SOFTWARE

1 – No spinners

7 IDEALS
FOR Local-First SOFTWARE

1 – No spinners

2 – Your work is not trapped on one device

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3 – The network is optional

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FOR Local-First SOFTWARE

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2 – Your work is not trapped on one device

3 – The network is optional

4 – Seamless collaboration

5 – The Long Now

6 – Security and privacy by default

7 – You retain ultimate ownership and control

Local-first ecosystem





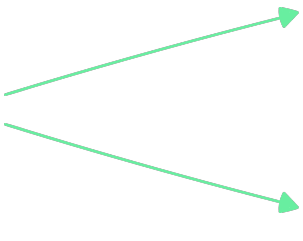




Evolu

Evolu

SQLite. CRDTs. Encrypted sync.



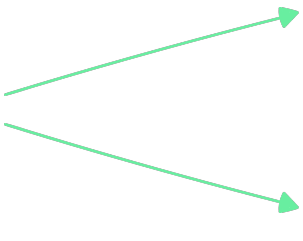
Local-first platform

TypeScript library

Focus on quality, maintainability, scaling

Evolu

SQLite. CRDTs. Encrypted sync.



Local-first platform

TypeScript library

Focus on quality, maintainability, scaling

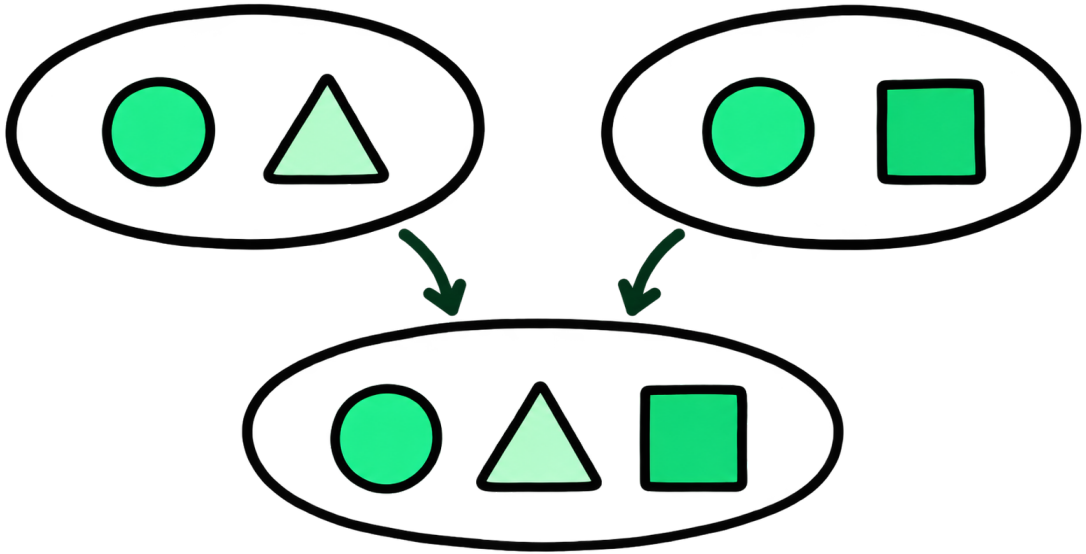
CRDTs

Conflict-free Replicated Data Types

G-Set

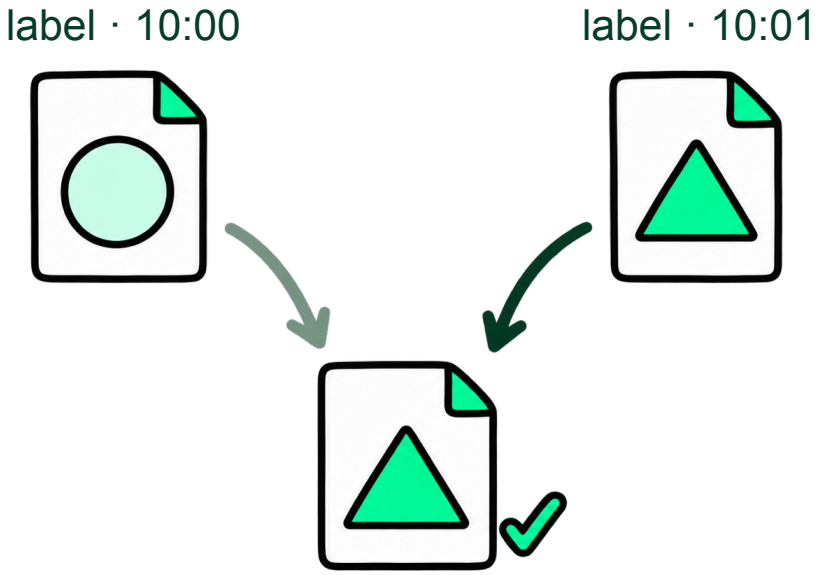
Grow-only Set

- Add
- Read
- Merge



LWW-Map

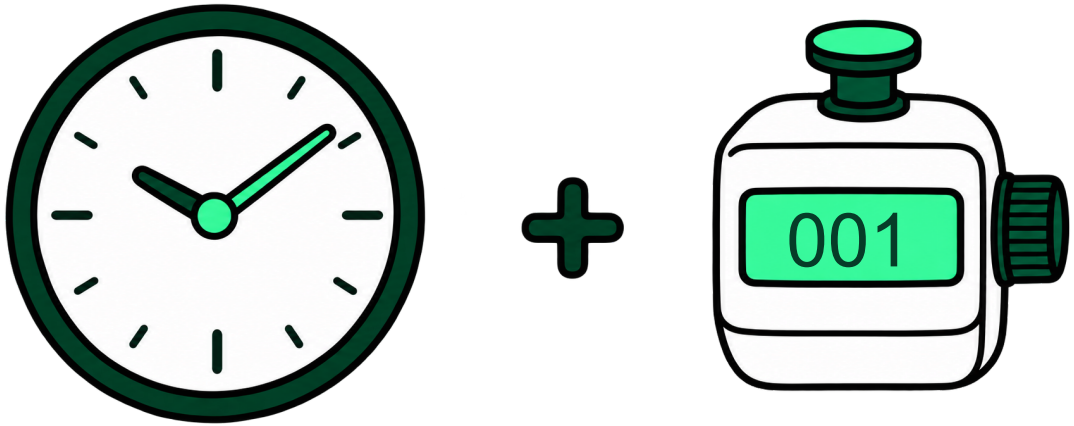
Last Write Wins - Map



Highest timestamp wins per key.

HLC

Hybrid Logical Clock

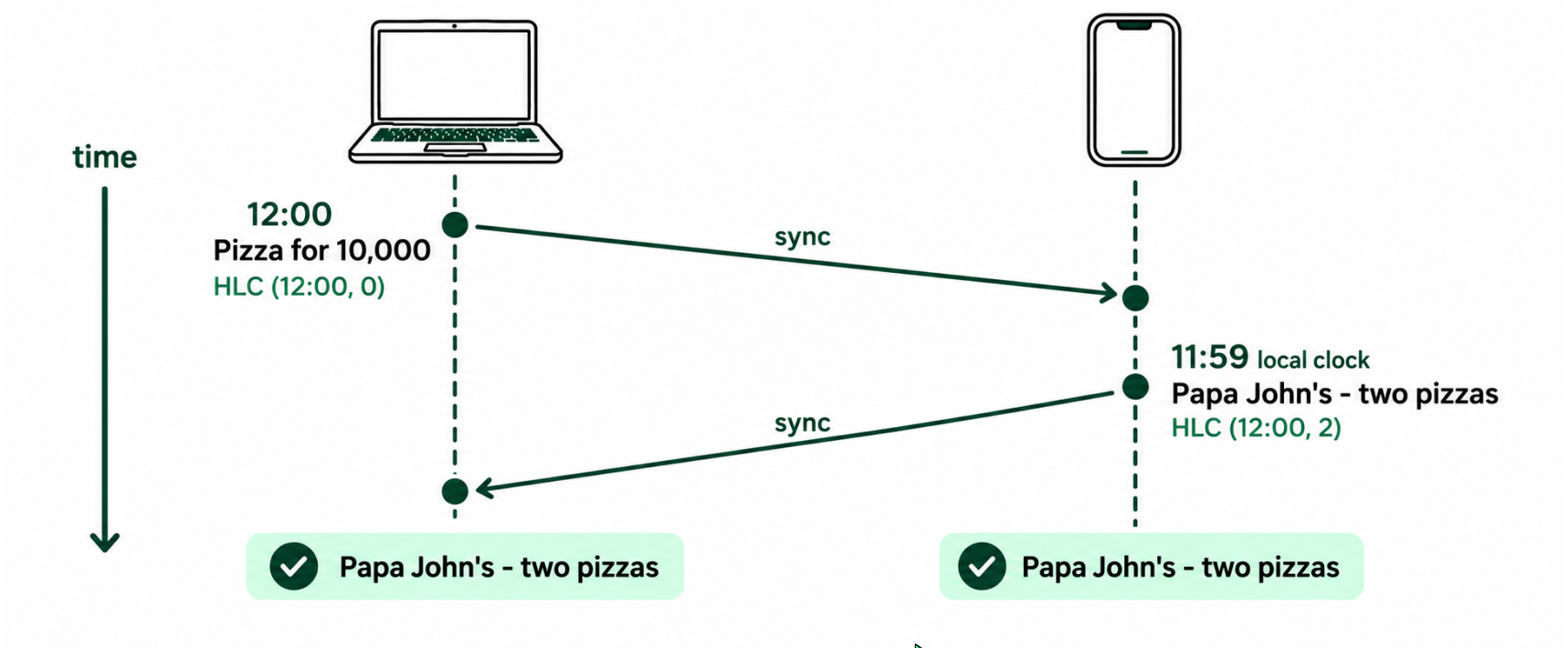


Physical time

Logical counter

(10:00, 0) → (10:00, 1) → (10:01, 0)

Eventual consistency

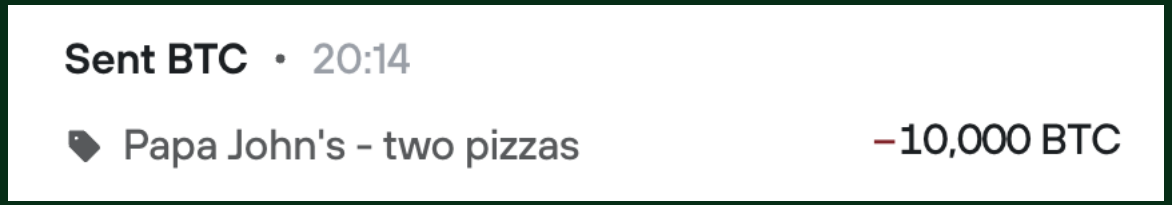


same set of operations ➡ same state

maybe in a different order!

Local change history

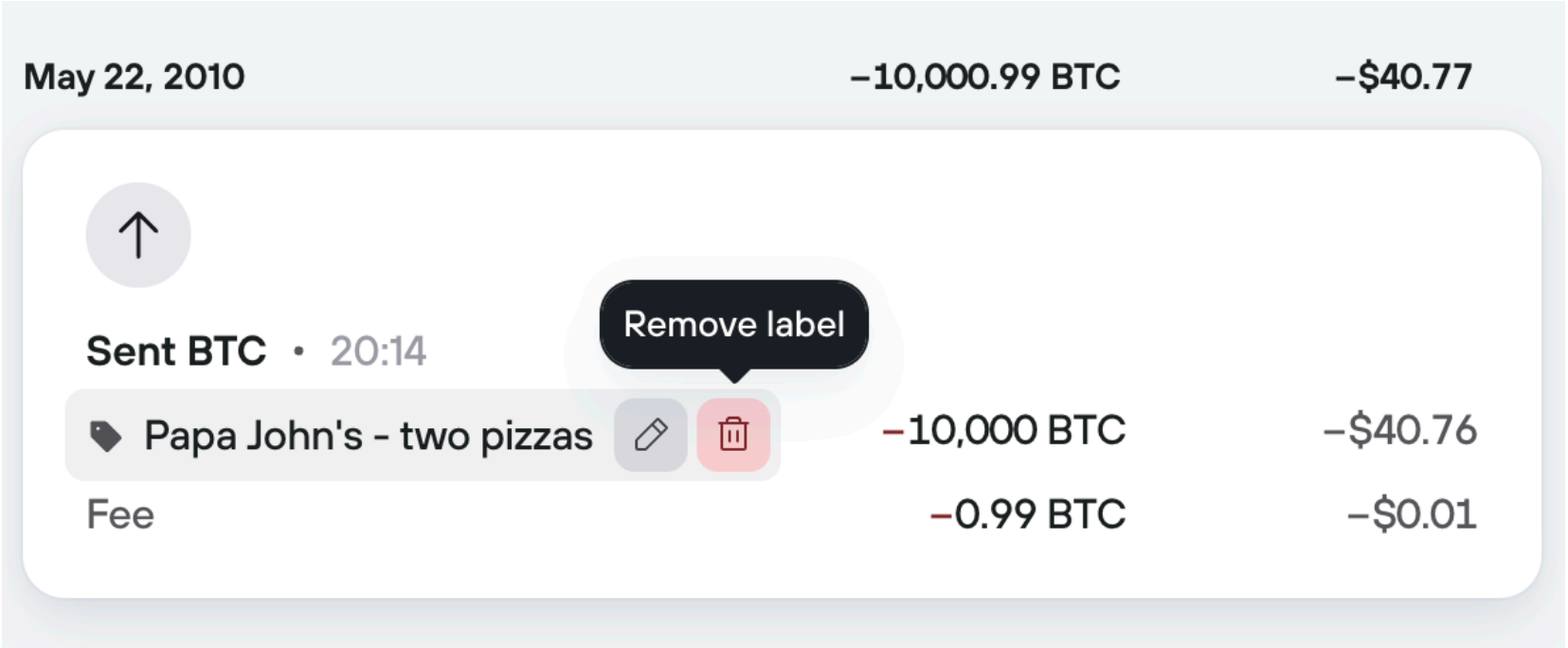
ownerId	table	id	column	timestamp	value
CD9BE586…	output	edacbf...3f7-0	label	2010-05-22T12:00:00.000Z 0bf5807cf26bdf138	**Pizza for 10,000**
CD9BE586…	output	edacbf...3f7-0	label	2010-05-22T12:00:00.000Z 0dcbd3ad32eca3c	**Papa John's - two pizzas**



Evolu Relay DB

ownerId	timestamp	change
CD9BE586...C8F	2010-05-22T12:00:00.000Z 0bf5807cf26bdf138	1C08C8A05557F3AA55DAED4D48437B4A130D435E4B46FB1BC00102BBB848E3DC3AC58D6FF5E7D3C2CF17E00973EAC09FAB32E766EA2824EABD1C32C35888D3E51FD3FD02EBC0DFD3E0A2559E12A5D7BFCF037087C7264ADCC7E4DB3F3014EFEBD37868311D6762ABE20441D301E560CFF22E7677D07CF425C462...
CD9BE586...C8F	2010-05-22T12:00:00.000Z 0dcbd3ad32eca3c	B8A476E4840C7C127A670F40DC844C10B2D3FD4CCF55512BD00189C0FB25A75FB28A94A410565F8CA003E4B08129CA2E3E20C8C8E0FBD37EDCCB40D4B7E1752A24D11C9ACAB50E576338A0533EE9E5362780B90C77C7A5F0F8986D920B5A802E0B1D4E11AF32E118C614FE4706253B828A1A26D79D1C5972D73C62F...
CD9BE586...C8F	2010-05-22T16:44:29.572Z 0dcbd3ad32eaca3cc	9626281E5E6424E992419E23A347DEA0CCD4681835AC81C7D0012CA1504CE406B207F51B000AB43091A748203E4F5E5DE078EE28291CC0DB0D3E67394495E9BE4995419951D2F88C3F6350156BF4B15958256D9D2296EE1ABE30EAE41571215486E544837B3ACEF13D5A3DB2086DEADA0E1CB7F4A7877B7BA89C9332...
EF80BCD5...CEF	2010-05-22T16:39:25.059Z 0ff4596ad00ca338a	08020E1DD8F9F3DE33843B4ABB6519A6050A1E86F19B70329002A07DD819F0BCC33D58CED48478353A6DC73449B8FA2107CAF66296530227966746CDEEDABD11BC1447D0DDF492A33B731C6FC91712BCFEACBAC90853308FC7661091FD7EAB7DD72683F176977254E3336847B9510277B61A69148CF0969DFAC5AD2...

Delete label



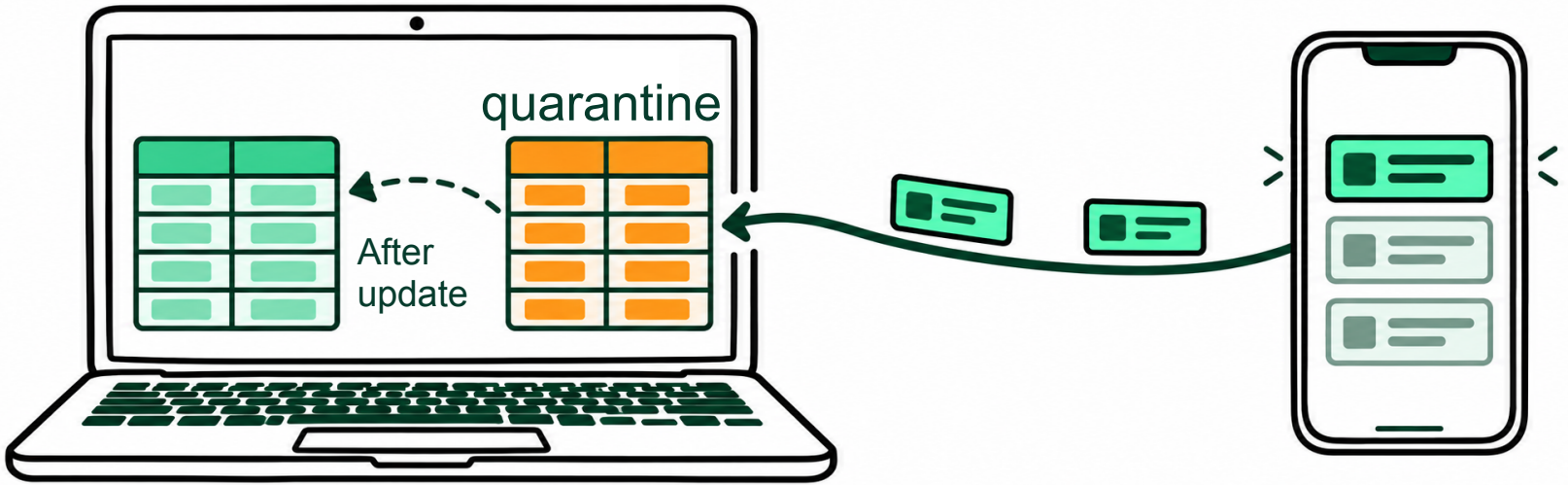
Local evolu DB - deletion

ownerId	table	id	column	timestamp	value
CD9BE586…	output	edacbf...3f7-0	label	2010-05-22T12:00:00.000Z 0bf5807cf26bdf138	**Pizza for 10,000**
CD9BE586…	output	edacbf...3f7-0	label	2010-05-22T12:00:00.000Z 0dcbd3ad32eca3c	**Papa John's - two pizzas**
CD9BE586…	output	edacbf...3f7-0	label	2026-10-22T12:00:00.000Z 0dcbd3ad32eca3c	

Old clients come back

Last year's app

Today's app

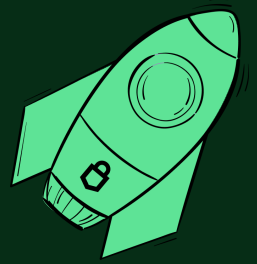


Waiting for an app update

Your servers are gone

Read.
Modify.
Export.



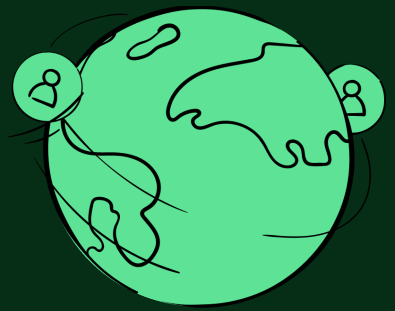
TOOLS ARE  READY

INCENTIVES ARE CHANGING



Apps I want to use

Local-first: lofi.so
Evolu: evolu.dev



Bring me your use case during the break

Matěj Kříž / Trezor Suite